SIAM AREFIN

Email | GitHub | LinkedIn | Portfolio

Research Interests: AI for Healthcare and Agriculture, Computer Vision, and Large Language Model.

Education

Shahjalal University of Science and Technology, Bangladesh Mar 2022 – Jul 2026
Bachelor of Science in Software Engineering ***CGPA: 3.10/4.00***

Publications

- *Siam Arefin, Md. Farzin Hossen, Fazle Rabbi Rakib*, "LLDDet: Extending a Lemon Leaf Disease Classification Dataset for Object Detection and Benchmarking with YOLOv11," *Manuscript in Preparation*, 2026.
- Joy Prokash Debnath, Kabir Hossen, Md. Sayeam Khandaker, Shawon Majid, Md. Mehrajul Islam, **Siam Arefin**, Preonath Chondrow Dev, Saifuddin Sarker, Tanvir Hossain, "omicML: An Integrative Bioinformatics and Machine Learning Framework for Transcriptomic Biomarker Identification," *bioRxiv*, 2025.
  Preprint: doi:10.1101/2025.10.25.684517

Research Experiences

LLDDet: Extending a Lemon Leaf Disease Classification Dataset for Object Detection and Benchmarking with YOLOv11

Undergraduate Thesis — Supervisor: Fazle Rabbi Rakib

- Extended a lemon leaf classification dataset into LLDDet, a 9-class object detection benchmark with bounding-box annotations for 1,352 images.
- Benchmarked YOLOv8m, YOLOv9m, YOLOv11m, and Faster R-CNN; **YOLOv11m achieved 0.93 test** mAP_{50}.
- Demonstrated disease-region localization under a resource-constrained 16 GB GPU environment.

omicML: An Integrative Bioinformatics and Machine Learning Framework for Transcriptomic Biomarker Identification

Machine Learning Research Engineer | Collaboration with Department of Biochemistry and Molecular Biology (BMB), SUST , Supervisor: Tanvir Hossain

- Developed **omicML**, a GUI integrating transcriptomic preprocessing, differential expression analysis, annotation, and ML.
- Implemented dataset integration, batch-effect correction, classifier benchmarking, and feature-importance analysis.
- Identified a **six-gene diagnostic model** for Mpox discrimination from other viral infections.

Professional Experience

Kite Games Studio Nov 2025 – Jan 2026

Software Engineer (ML Team) Intern | Mohakhali DOHS, Dhaka — Onsite

- Conducted R&D in image processing, machine learning, and generative AI for production-oriented applications.
- Worked with Diffusion and FLUX-based generative AI models for image generation and processing.
- Contributed to an Android application with 5M+ downloads and 700K+ daily active users.

Academic and Competitive Achievements

- **1st Place – Multi-Instance Object Detection Challenge**, Duality AI International Competition; developed YOLO-based detection with ensemble learning and Sim2Real adaptation, achieving a **0.995 private leaderboard score**.
- **1st Runner-up – ML Olympiad**, CO_2 Emissions Prediction Challenge; developed an ML pipeline involving preprocessing, feature engineering, model training, and optimization.
- **4th Place – Robi Datathon 3.0 Qualifier, Problem 1**, Robi; developed a purchase prediction solution using historical customer transaction data and product-level information.
- **4th Place – DL Enigma 1.0**, SUST CSE Carnival 2024; developed and optimized a YOLOv11m model for vehicle detection.
- **5th Place – Geospatial Object Detection Challenge**, Duality AI & LunateAI; trained aerial object detection models using 100% synthetic data and applied Sim2Real adaptation for real-world rural building detection, ranking **5th among 54 teams**.
- **Top 10 – DL Sprint 3.0**, Bengali AI Math Olympiad; developed prompt-engineering-based solutions for Bengali mathematical problems using the Qwen model.
- **Top 10 – Bhashamul: Bengali Regional IPA Transcription**, UIU Computer Club, UIU CSE & Bengali.AI; developed an NLP/deep learning solution for Bengali regional dialect text-to-IPA transcription, contributing to low-resource Bengali language technology.
- **800+ Problems Solved** on Codeforces and other competitive programming platforms, covering algorithms, data structures, and problem solving.

Projects

Multi-Instance Object Detection GitHub

- Merged synthetic datasets from multiple sources and applied ensemble learning to reduce the Sim2Real domain gap.
- Implemented custom post-processing using NMS and confidence filtering and fine-tuned YOLO models to improve generalization to real-world images.

HealthMonitor – AI-Powered Health Monitoring App GitHub

- Developed and deployed a MERN-stack application with a custom-trained machine learning model for cardiovascular disease prediction.
- Integrated the Google Gemini API for intelligent advice and developed a RAG-based system for personalized insights using medical records and health-related data.

AI Processing API – AI/ML Backend System GitHub

- Designed a modular AI backend using layered architecture, Provider/Adapter pattern, and Dependency Injection.
- Built REST APIs for audio transcription and medical-document OCR with validation, structured extraction, testing, and Dockerized deployment.

Volunteer Experience

Tabligh Volunteer & Jamat Leader 2025–2026

- Completed three periods of community volunteer service and served as a **Jamat Leader**.
- Engaged with diverse individuals across **Dhaka, Sylhet, Kishoreganj, Madaripur, and Chapainawabganj**, developing practical experience in **human behavior, communication, persuasion, and leadership**.

Programming Language

Python, C, C++, Java, JavaScript, SQL